

1. Atomic Structure

Atomic Number (Z) = Number of protons

Mass Number (A) = Number of protons + neutrons

Number of neutrons = A - Z

Electron configuration: follows Aufbau principle, Hund's rule, Pauli's exclusion principle

2. Mole Concept

Moles, $n = \text{Mass (m)} / \text{Molar mass (M)}$

Moles, $n = \text{Number of particles} / \text{Avogadro's number } (6.022 \times 10^{23})$

Gas molar volume at STP = 22.4 L/mol

Ideal Gas Equation: $PV = nRT$

Concentration: $C = n / V \text{ (mol/L)}$

3. Chemical Bonding

Ionic bond: formed by transfer of electrons

Covalent bond: formed by sharing of electrons

Polar covalent: unequal sharing of electrons

Electronegativity difference (ΔEN) $\rightarrow$ determines bond type:

$\Delta EN > 1.7 \rightarrow$ Ionic

$0.4\text{--}1.7 \rightarrow$ Polar covalent

$< 0.4 \rightarrow$ Nonpolar covalent

Bond order (BO) = (Number of bonding electrons – Number of antibonding electrons) / 2

4. Stoichiometry

$n = m/M$ (mass to moles)

$n = C \times V$ (solution concentration)

Empirical formula: simplest whole number ratio of atoms

Molecular formula = Empirical formula $\times$ (Molecular mass / Empirical formula mass)

5. Thermodynamics

$\Delta H = H_{\text{products}} - H_{\text{reactants}}$

$\Delta G = \Delta H - T\Delta S$

Spontaneous if $\Delta G < 0$

Hess's Law: Total enthalpy change = sum of enthalpy changes of individual steps

6. Electrochemistry

Cell potential: $E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$

Faraday's laws: $m = (Q \times M) / (n \times F)$, $Q = I \times t$

F (Faraday constant) = 96485 C/mol e^-

7. Chemical Kinetics

Rate = $k [A]^m [B]^n$

Half-life for first order: $t_{1/2} = 0.693 / k$

Half-life for second order: $t_{1/2} = 1 / (k[A]_0)$

8. Equilibrium

$K_c = [\text{products}]^p / [\text{reactants}]^r$

$K_p = (P_{\text{products}})^p / (P_{\text{reactants}})^r$

$K_p = K_c(RT)^{\Delta n}$, Δn = moles of gas products - moles of gas reactants

Le Chatelier's Principle: System shifts to oppose change

9. Acids & Bases

$$\text{pH} = -\log[\text{H}^+]$$

$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{pH} + \text{pOH} = 14$$

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}, K_b = \frac{[\text{BH}^+][\text{OH}^-]}{[\text{B}]}$$

$$K_w = [\text{H}^+][\text{OH}^-] = 1 \times 10^{-14} \text{ at } 25^\circ\text{C}$$

10. Common Organic Chemistry

General formula: Alkanes: $\text{C}_n\text{H}_{2n+2}$, Alkenes: C_nH_{2n} , Alkynes: $\text{C}_n\text{H}_{2n-2}$

Functional groups: $-\text{OH}$ (alcohol), $-\text{COOH}$ (carboxylic acid), $-\text{NH}_2$ (amine)

Isomerism: Structural, Geometrical, Optical